Research Statement

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Towards Reliable and Efficient Knowledge-Centric Reasoning in Large Language Models

Large language models (LLMs) have evolved from standalone language generators into increasingly capable reasoning systems and autonomous agents. By interacting with external tools, retrieving information, and maintaining memory across multiple steps, these systems can tackle complex tasks that extend beyond their parametric knowledge. However, access to more information does not necessarily lead to better reasoning. LLMs often struggle to identify useful evidence, become distracted by accumulated context, and fail to consistently incorporate newly acquired knowledge into their decisions.

My research is motivated by a fundamental question: **How can language models acquire, organize, and utilize external knowledge to support reliable and efficient reasoning?**

Throughout my research, I have approached this question from three interconnected perspectives: refining retrieved evidence, organizing knowledge across iterative reasoning, and improving the efficiency of information utilization. Building upon these experiences, my long-term vision is to develop knowledge-centric LLMs and agents that can adaptively determine what information they need, maintain reliable and evolving memories, and reason efficiently over essential knowledge.

1 Overview of Previous Work

1.1 Refining Knowledge for Reliable Retrieval-Augmented Generation

How can language models distinguish useful knowledge from noisy retrieved evidence? Retrieval-augmented generation (RAG) enhances language models by incorporating external information, but retrieved documents frequently contain irrelevant, redundant, or misleading evidence. Models may fail to distinguish useful information from noise, limiting the reliability of their reasoning.

To address this challenge, I proposed **RankCoT** (Wu et al., 2025), a knowledge refinement framework that enables language models to rank and refine retrieved information before generating answers. Rather than directly consuming retrieved documents, RankCoT trains models to identify and prioritize relevant evidence through ranking-based reasoning. This work established my initial research interest in understanding how models can evaluate the utility of external information instead of passively accepting retrieved context.

I subsequently extended this perspective to situations where acquiring additional information incurs explicit costs. In **Cost-Aware RAG** (Wu et al., 2026b), I co-developed a framework that balances the expected utility of additional evidence against its acquisition cost. This research highlighted that effective knowledge acquisition requires not only identifying relevant information but also deciding whether further retrieval is worthwhile.

1.2 Organizing Knowledge across Long-Horizon Reasoning

How can language models maintain access to essential knowledge as reasoning contexts grow? In complex tasks involving iterative retrieval, language models progressively accumulate information. Although such information may be useful, increasing context length can distract models from previously acquired evidence and reduce their ability to locate relevant knowledge.

Motivated by this challenge, I proposed **KAIR** (Wu et al., 2026a), a knowledge-anchoring framework that constructs evolving knowledge indices throughout iterative retrieval and answer generation. Rather than repeatedly processing an expanding context without explicit organization, KAIR provides structured indices that help models locate and utilize important accumulated information.

My interest in long-horizon reasoning also extends to autonomous agents. In **EigentSearch-Q+** (Zhang et al., 2026), I contributed to structured tools for query planning, research-progress monitoring, and evidence extraction in deep-research agents. I also developed an evaluation framework supporting multiple knowledge-intensive benchmarks[1]. Together, these experiences motivated a broader perspective: **effective reasoning requires not only acquiring knowledge but also maintaining structured and accessible representations of information over time**.

1.3 Improving the Efficiency of LLM Reasoning

Can language models reason more efficiently without losing essential intermediate information? As reasoning models generate increasingly long trajectories, substantial computation may be spent on redundant intermediate steps. This raises the question of how to preserve critical reasoning information while reducing unnecessary computation. In **Reasoning Compression with Mixed-Policy Distillation** Yang et al. (2026), I collaborated on a framework that transfers reasoning behaviors from multiple teacher-generated policies into a more efficient student model. I participated in method implementation and experimental evaluation, investigating how reasoning behavior can be compressed while maintaining task performance.

This experience encouraged me to look beyond textual reasoning compression and explore **how essential information can be identified and preserved within the internal representations of language models**.

2 Future Research Vision

My future research aims to develop **adaptive, reliable, and computationally efficient knowledge management mechanisms for LLM reasoning and autonomous agents**. I am particularly interested in three complementary directions.

2.1 Adaptive Knowledge Acquisition and Sufficiency Verification

Current retrieval-augmented systems often rely on fixed retrieval strategies or loosely defined uncertainty signals to decide whether additional evidence is necessary. Consequently, they may repeatedly retrieve information even when existing evidence is sufficient, or prematurely stop searching before critical knowledge has been acquired. I aim to investigate **how LLMs can explicitly evaluate the sufficiency of accumulated evidence for a reasoning task**. One possible approach is to assess whether available knowledge supports the essential claims and dependencies required for a solution. Models could then determine when further retrieval is necessary and when existing information should be reused.

2.2 Reliable Memory Construction and Revision

As autonomous agents interact with external environments, they increasingly rely on persistent memories containing accumulated facts, experiences, and reasoning strategies. However, these memories may become outdated, contradictory, or contaminated by unreliable information. My goal is to study **how agents can construct, evaluate, and revise persistent memories while maintaining knowledge consistency and source reliability**.

I envision memory systems that distinguish different types of knowledge, preserve evidence sources, and adapt their representations based on newly acquired information and feedback. Rather than merely appending information, agents should be able to identify contradictions, revise previous beliefs, and reconsider the reliability of stored knowledge. An important extension is to investigate robustness against misleading or adversarial evidence. I am interested in understanding how errors or manipulated information propagate through persistent memory and subsequently influence agent behavior.

2.3 Reasoning-Aware Memory Compression and Information Preservation

Beyond external memory, LLMs also maintain substantial internal contextual information through their attention mechanisms and KV-cache. As reasoning trajectories grow longer, retaining all previous information becomes computationally expensive and may introduce unnecessary contextual interference. My future research aims to explore adaptive and efficient information management mechanisms for LLM

[1] `https://github.com/camel-ai/eigent_search`

reasoning, with a particular interest in understanding how models can better balance information preservation, computational efficiency, and reasoning performance. Ultimately, I hope to develop more efficient and scalable reasoning systems.

3 Final Remarks

My previous research has progressed from **refining retrieved evidence** to **organizing accumulated knowledge**, and more recently to **improving reasoning efficiency**. These experiences have shaped my belief that reliable LLM reasoning depends on the ability to selectively acquire, maintain, revise, and utilize information rather than simply processing increasingly large contexts.

During my PhD, I hope to deepen my understanding of the mechanisms underlying knowledge-intensive reasoning and develop methods that improve both the reliability and efficiency of autonomous language model systems. Ultimately, I envision LLM agents that can reason not only over what they know, but also about **what they need to know, what information they should trust, and what knowledge they should preserve or revise**.

References

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